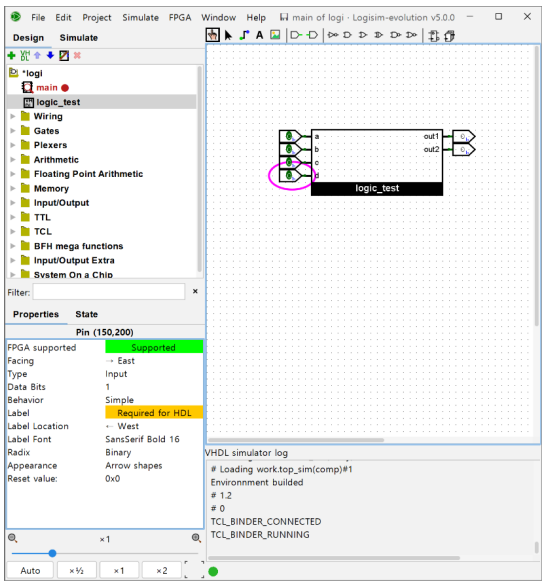
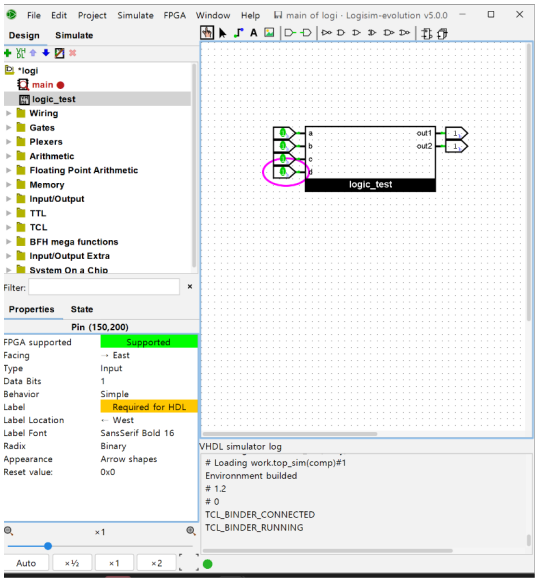
VHDL-assignment 2

2022042251김동한

1. AND, OR





-작성한코드

```
library ieee;
use ieee.std_logic_1164.all;

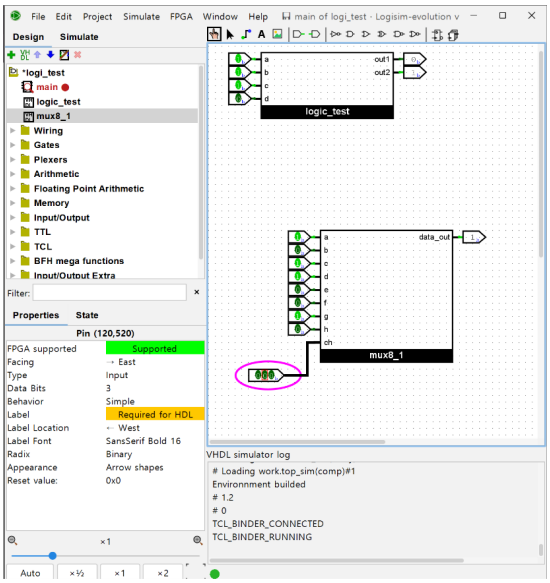
entity logic_test is
    port (
```

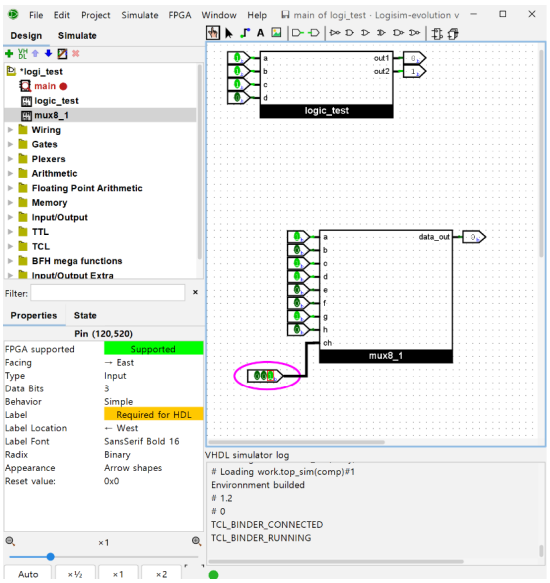
```
        a, b, c, d : in std_logic;
        out1       : out std_logic;
        out2       : out std_logic
    );
end entity logic_test;

architecture bhv of logic_test is
begin
    out1 <= a and b and c and d;
    out2 <= a or b or c or d;
end architecture bhv;
```

>>> 입력이 모두 1일 때 AND 출력이 1이고, 하나라도 1이면 OR 출력이 1이 되는 것을 확인했다.

2. 8-to-1 MUX





-작성한코드

```
library ieee;
use ieee.std_logic_1164.all;

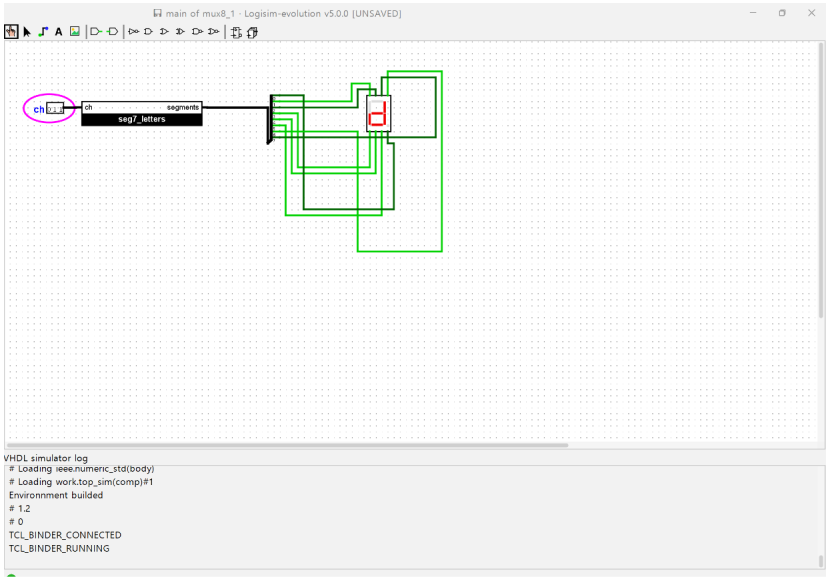
entity mux8_1 is
    port (
        a, b, c, d, e, f, g, h : in std_logic;
        ch                     : in std_logic_vector(2
            downto 0);
        data_out               : out std_logic
```

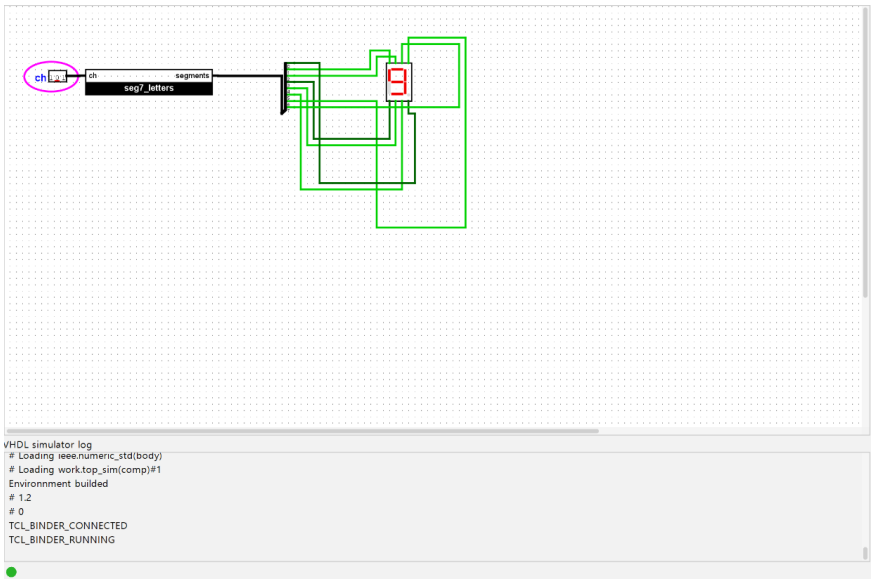
```
    );
end entity mux8_1;

architecture bhv of mux8_1 is
begin
    process(a, b, c, d, e, f, g, h, ch)
    begin
        case ch is
            when "000" => data_out <= a;
            when "001" => data_out <= b;
            when "010" => data_out <= c;
            when "011" => data_out <= d;
            when "100" => data_out <= e;
            when "101" => data_out <= f;
            when "110" => data_out <= g;
            when "111" => data_out <= h;
            when others => data_out <= '0';
        end case;
    end process;
end architecture bhv;
```

>>> ch=000에서는 a가, ch=001에서는 b가 출력된다. 선택 신호에 따라 입력 하나가 data_out으로 나오는 것을 확인했다.

3. 7-Segment





-작성한코드

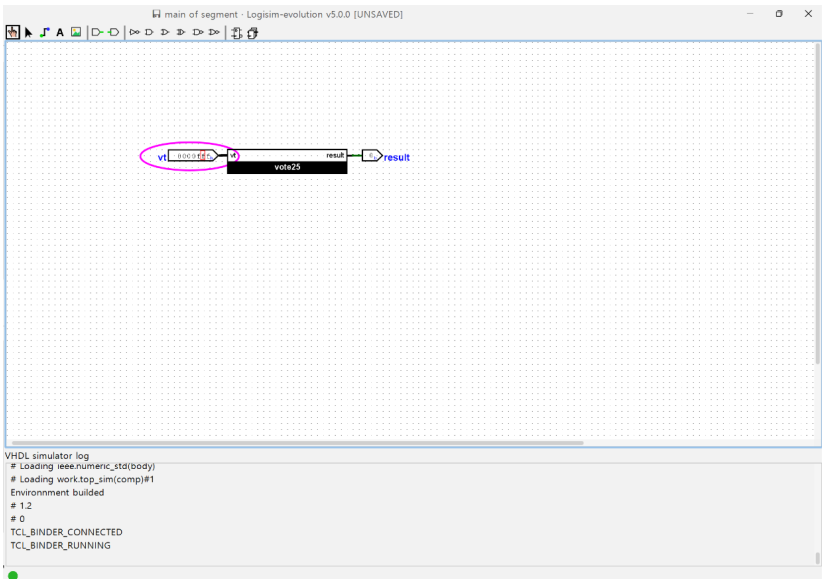
```
library ieee;
use ieee.std_logic_1164.all;

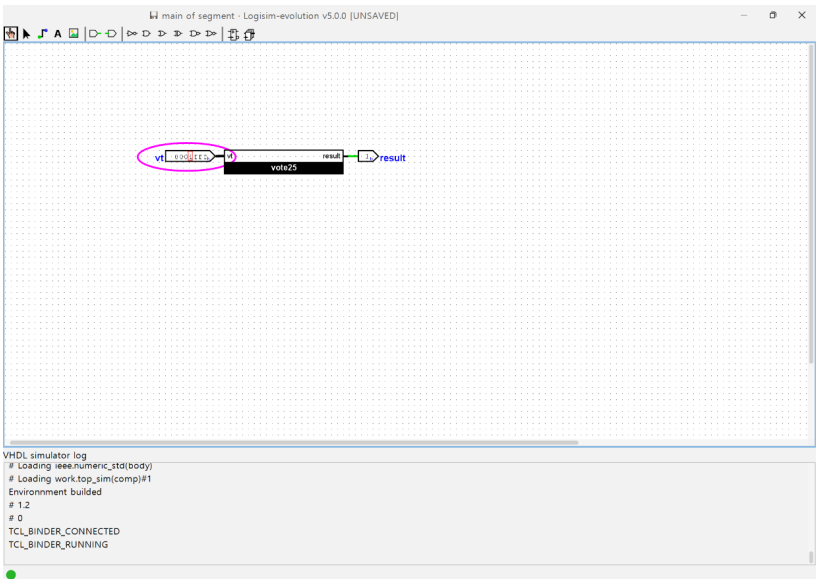
entity seg7_letters is
    port (
        ch       : in  std_logic_vector(2 downto 0);
        segments : out std_logic_vector(7 downto 0)
    );
end entity seg7_letters;
```

```
architecture bhv of seg7_letters is
begin
    process(ch)
    begin
        case ch is
            when "000" => segments <= "11111010"; -- a
            when "001" => segments <= "00111110"; -- b
            when "010" => segments <= "00011010"; -- c
            when "011" => segments <= "01111010"; -- d
            when "100" => segments <= "10001110"; -- f
            when "101" => segments <= "11110110"; -- g
            when others => segments <= "00000000";
        end case;
    end process;
end architecture bhv;
```

>>> ch 값에 따라 a, b, c, d, f, g가 표시되도록 했다. 위 화면은 d와 g를 표시한 결과다.

4. 25-person Voting Machine





-작성한코드

```
library ieee;
use ieee.std_logic_1164.all;

entity vote25 is
    port (
        vt     : in  std_logic_vector(24 downto 0);
        result : out std_logic
    );
end entity vote25;
```

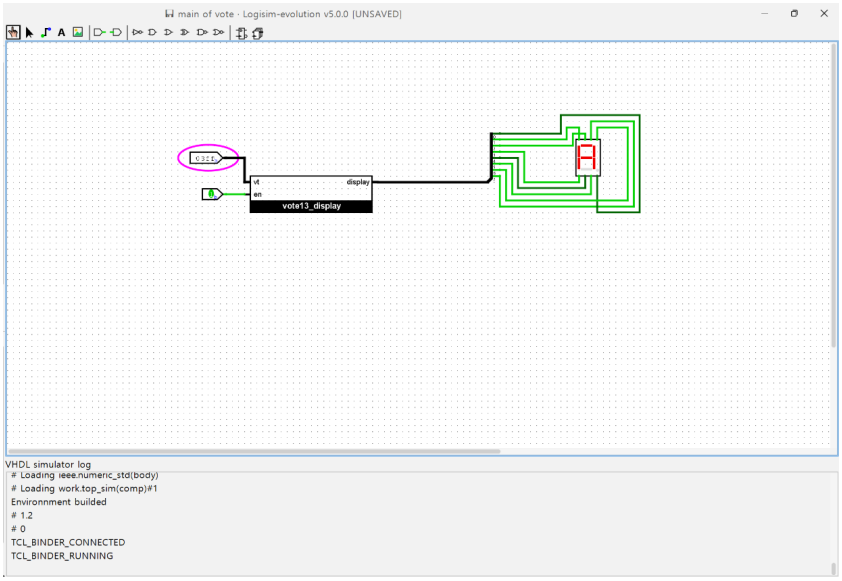
```
architecture bhv of vote25 is
begin
    process(vt)
        variable vote_count : integer range 0 to 25;
    begin
        vote_count := 0;

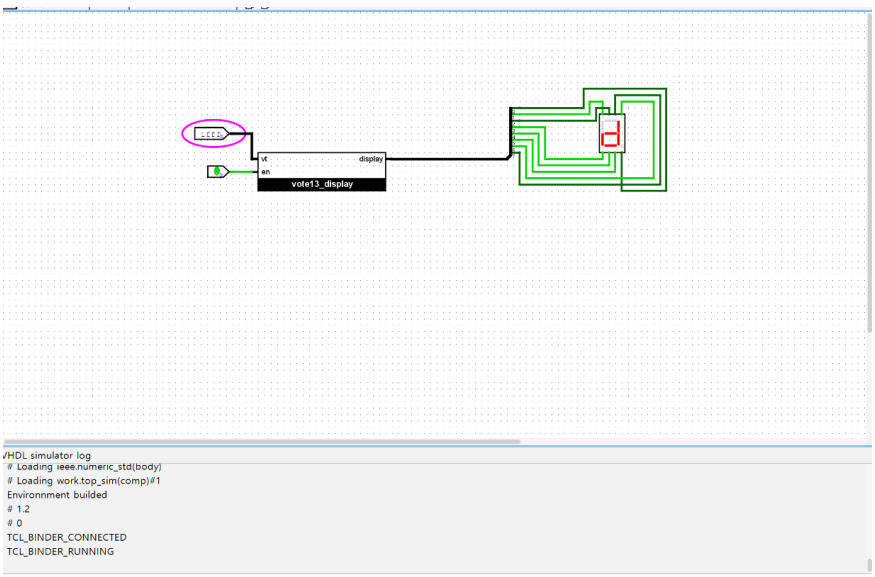
        for i in vt'range loop
            if vt(i) = '1' then
                vote_count := vote_count + 1;
            end if;
        end loop;

        if vote_count >= 13 then
            result <= '1';
        else
            result <= '0';
        end if;
    end process;
end architecture bhv;
```

>>> 0000FFF는 찬성 12명이라 출력이 0이고, 0001FFF는 찬성 13명이라 출력이 1이다. 13명 이상이면 통과한다.

5. 13-bit Voting Count





-작성한코드

```
library ieee;
use ieee.std_logic_1164.all;

entity vote13_display is
    port (
        vt      : in  std_logic_vector(12 downto 0);
        en      : in  std_logic;
        display : out std_logic_vector(7 downto 0)
    );
```

```
end entity vote13_display;

architecture bhv of vote13_display is
begin
    process(vt, en)
        variable vote_count : integer range 0 to 15;
    begin
        vote_count := 0;

        for i in vt'range loop
            if vt(i) = '1' then
                vote_count := vote_count + 1;
            end if;
        end loop;

        if en = '0' then
            display <= "00000000";
        else
            -- Bit order: a b c d e f g dp
            case vote_count is
                when  0 => display <= "11111100"; -- 0
                when  1 => display <= "01100000"; -- 1
                when  2 => display <= "11011010"; -- 2
                when  3 => display <= "11110010"; -- 3
                when  4 => display <= "01100110"; -- 4
                when  5 => display <= "10110110"; -- 5
                when  6 => display <= "10111110"; -- 6
                when  7 => display <= "11100000"; -- 7
                when  8 => display <= "11111110"; -- 8
                when  9 => display <= "11110110"; -- 9
                when 10 => display <= "11101110"; -- A
                when 11 => display <= "00111110"; -- b
                when 12 => display <= "10011100"; -- C
                when 13 => display <= "01111010"; -- d
                when 14 => display <= "10011110"; -- E
                when 15 => display <= "10001110"; -- F
                when others => display <= "00000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

>>> 찬성 10명일 때 A, 13명일 때 d가 표시된다. en=0이면 표시가 꺼지고 en=1이면 결과가 나온다. 입력이 13비트라 14와 15는 나올 수 없다.